

Polyglot Persistence

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R00225669

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Research on Integrating MongoDB cluster with Neo4j with the Latest Version of DocManager

Option 2

Task: Write a short report to document the steps of setting up doc manager with neo4j. In the report, you need to indicate the version of your software and the operating systems you used. In the report, you need to prove that the connection is functioning with the diabetes self-deployed cluster example. (You need to answer three queries in the spec).

Operating System – Windows Version 10,0,26100,4061

Neo4j Version – desktop 5.24.0

MongoDB Version – 8.0 Community

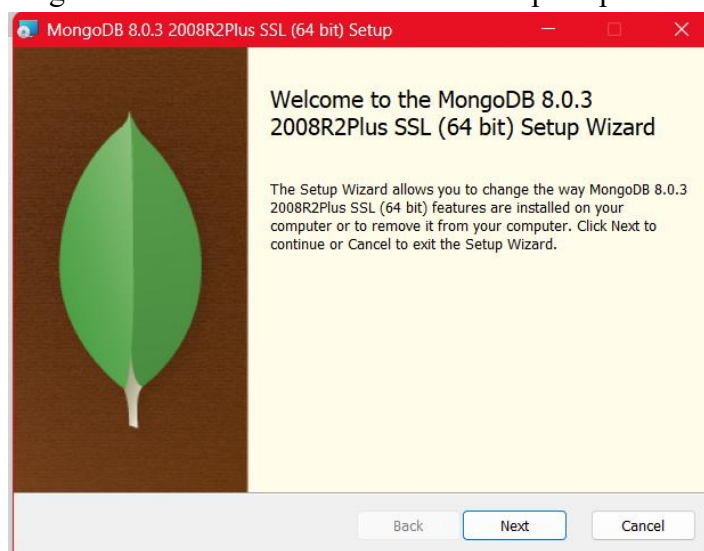
APOC Version 5.24.0

Mongo-connector – 3,2,0

Neo4j_doc_manager – 0.5.1

Installing MongoDB

First you need to make sure that you have MongoDB running on the computers. You will need the MongoDB server (mongod.exe) and the shell installed. To download this is simple just go to the MongoDB Download Centre, and download it to your machine, the only thing is that you have to make sure to add MongoDB to your system PATH as this is what lets you run MongoDB commands from the command prompt without typing the full path.



Create the Data Directory

MongoDB stores its data files in a directory. It automatically looks for C:\data\db, but if this folder doesn't exist, MongoDB will show an error when you try to start it.

It's possible that you might have to import the data, to do this we use the mongoimport

```
Command Prompt
Microsoft Windows [Version 10.0.26100.4061]
(c) Microsoft Corporation. All rights reserved.

C:\Users\rld>mkdir C:\data\db
A subdirectory or file C:\data\db already exists.

C:\Users\rld>
```

Starting MongoDB

Next, you'll need to start the MongoDB server process so that it can listen for connections to store data. Drag and drop your mongod.exe "C:\Program Files\MongoDB\Server\8.0\bin\mongod.exe" --dbpath "C:\data\db" and hit enter. Minimise this and let it run in the background. If you close it, MongoDB will stop running.

```
Command Prompt - "C:\Progr
Microsoft Windows [Version 10.0.26100.4061]
(c) Microsoft Corporation. All rights reserved.

C:\Users\rld>"C:\Program Files\MongoDB\Server\8.0\bin\mongod.exe"
{"t":{"$date":"2025-05-26T03:47:01.002+01:00"},"s":"I", "c":"CONTROL", "id":23285, "ctx":"thread1","msg":"A
lly disabling TLS 1.0, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2025-05-26T03:47:01.008+01:00"},"s":"I", "c":"CONTROL", "id":5945603, "ctx":"thread1","msg":"H
ading initialized"}
{"t":{"$date":"2025-05-26T03:47:01.013+01:00"},"s":"I", "c":"NETWORK", "id":4648601, "ctx":"thread1","msg":"T
CP FastOpen unavailable. If TCP FastOpen is required, set at least one of the related parameters","attr":{"rela
ters":["tcpFastOpenServer","tcpFastOpenClient","tcpFastOpenQueueSize"]}}
{"t":{"$date":"2025-05-26T03:47:01.026+01:00"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"thread1","msg":"I
d wire specification","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":25},"incom
lClient":{"minWireVersion":0,"maxWireVersion":25},"outgoing":{"minWireVersion":6,"maxWireVersion":25},"isInter
:true}}}}
{"t":{"$date":"2025-05-26T03:47:01.037+01:00"},"s":"I", "c":"TENANT_M", "id":7091600, "ctx":"thread1","msg":"S
ent Migration Access Block Request"}

```

Connect to MongoDB

Open a new Command Prompt to start the MongoDB Shell and run mongosh, this will connect you to the running MongoDB server. You need this second terminal window to interact with MongoDB while the server is running in the first window.

```
mongosh mongodb://127.0.0.1
Microsoft Windows [Version 10.0.26100.4061]
(c) Microsoft Corporation. All rights reserved.

C:\Users\rld>C:\Users\rld\Downloads\mongosh-2.3.3-win32-x64\mongosh-2.3.3-win32-x64\bin\mongosh.exe
Current Mongosh Log ID: 68329f0fe2529933220d818f
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2
.3.3
Using MongoDB:      8.0.3
Using Mongosh:      2.3.3
mongosh 2.5.1 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2025-05-21T03:59:13.634+01:00: Access control is not enabled for the database. Read and write access to data and conf
iguration is unrestricted
-----
```

Importing the Diabetes Data

Download the diabetes_data.json file. In the MongoDB shell, create the database and import the data. This makes it so that you can use the diabetes dataset in the MongoDB.

```
2.7/2.7 MB 10.3 MB/s eta 0:00:00
Downloading pytz-2025.2-py2.py3-none-any.whl (509 kB)
Building wheels for collected packages: neo4j-doc-manager
  Building wheel for neo4j-doc-manager (pyproject.toml) ... done
  Created wheel for neo4j-doc-manager: filename=neo4j_doc_manager-0.1.1-py2.py3-none-any.whl size=8077 sha256=0aeea9a45834fe4cebcf3c8a8c8972d8c6e35ec1edb6aad88f6613345a162c52
  Stored in directory: c:\users\rlord\appdata\local\pip\cache\wheels\b
f\3d\83\7c52773a4c8df77535d94d403e2a4ba3394ab592ac44a933f2
Successfully built neo4j-doc-manager
Installing collected packages: pytz, monotonic, urllib3, six, pygments, pillow, packaging, idna, charset-normalizer, certifi, requests, pansi, interchange, py2neo, neo4j-doc-manager
Successfully installed certifi-2025.4.26 charset-normalizer-3.4.2 idna-3.10 interchange-2021.0.4 monotonic-1.6 neo4j-doc-manager-0.1.1 packaging-25.0 pansi-2024.11.0 pillow-11.2.1 py2neo-2021.2.4 pygments-2.19.1 pytz-2025.2 requests-2.32.3 six-1.17.0 urllib3-2.4.0

[notice] A new release of pip is available: 24.2 -> 25.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip

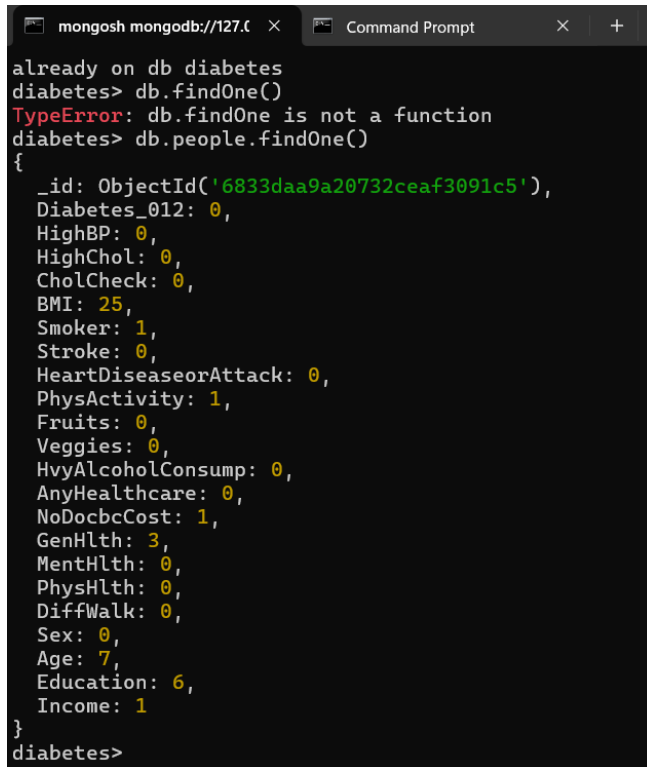
C:\Users\rlord>
C:\Users\rlord>
```

```
C:\Users\rlord>
C:\Users\rlord>mongo-connector -m localhost:27017 -t localhost:7474 -d
neo4j_doc_manager
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "C:\Users\rlord\AppData\Local\Programs\Python\Python312\Scripts\
mongo-connector.exe\__main__.py", line 4, in <module>
  File "C:\Users\rlord\AppData\Local\Programs\Python\Python312\Lib\sit
e-packages\mongo_connector\connector.py", line 38, in <module>
    from mongo_connector.command_helper import CommandHelper
  File "C:\Users\rlord\AppData\Local\Programs\Python\Python312\Lib\sit
e-packages\mongo_connector\command_helper.py", line 20, in <module>
    from mongo_connector.namespace_config import NamespaceConfig
  File "C:\Users\rlord\AppData\Local\Programs\Python\Python312\Lib\sit
e-packages\mongo_connector\namespace_config.py", line 18, in <module>
    from collections import namedtuple, MutableSet
ImportError: cannot import name 'MutableSet' from 'collections' (C:\Us
ers\rlord\AppData\Local\Programs\Python\Python312\Lib\collections\__in
it__.py)

C:\Users\rlord>
```

Check that your data is there

Just check that you have imported the data correctly. This also switches to your database and shows one document from your collection

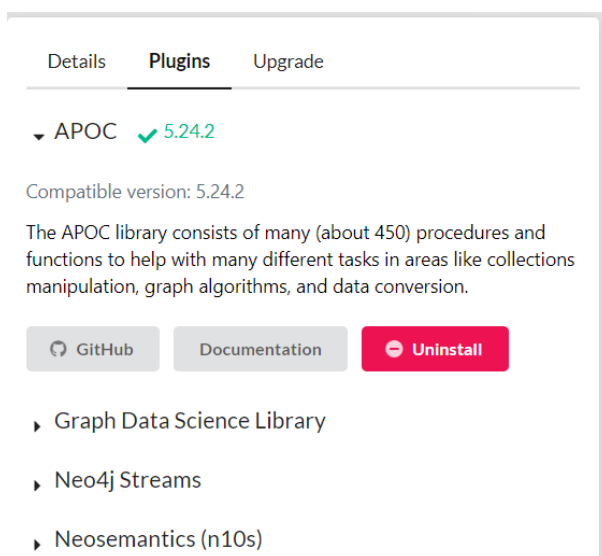


```
mongosh mongodb://127.0.0.1:27020/ > use diabetes
diabetes> db.findOne()
TypeError: db.findOne is not a function
diabetes> db.people.findOne()
{
  _id: ObjectId('6833daa9a20732ceaf3091c5'),
  Diabetes_012: 0,
  HighBP: 0,
  HighChol: 0,
  CholCheck: 0,
  BMI: 25,
  Smoker: 1,
  Stroke: 0,
  HeartDiseaseorAttack: 0,
  PhysActivity: 1,
  Fruits: 0,
  Veggies: 0,
  HvyAlcoholConsump: 0,
  AnyHealthcare: 0,
  NoDocbcCost: 1,
  GenHlth: 3,
  MentHlth: 0,
  PhysHlth: 0,
  DiffWalk: 0,
  Sex: 0,
  Age: 7,
  Education: 6,
  Income: 1
}
diabetes>
```

Setting Up Neo4j and APOC

Once MongoDB is up and running you can move onto Neo4j set up. Neo4j is a graph database, which means it's good at showing relationships. Download and install it from neo4j.com, to install it in Neo4j Desktop, to your database then Plugins and Install APOC.

```
C:\Users\rlord>C:\Users\rlord\Downloads\neo4j-community-5.26.6-windows\neo4j-community-5.26.6\bin\neo4j.bat console
Directories in use:
home:          C:\Users\rlord\Downloads\neo4j-community-5.26.6-windows\neo4j-community-5.26.6
config:        C:\Users\rlord\Downloads\neo4j-community-5.26.6-windows\neo4j-community-5.26.6\conf
logs:          C:\Users\rlord\Downloads\neo4j-community-5.26.6-windows\neo4j-community-5.26.6\logs
plugins:       C:\Users\rlord\Downloads\neo4j-community-5.26.6-windows\neo4j-community-5.26.6\plugins
import:        C:\Users\rlord\Downloads\neo4j-community-5.26.6-windows\neo4j-community-5.26.6\import
data:          C:\Users\rlord\Downloads\neo4j-community-5.26.6-windows\neo4j-community-5.26.6\data
certificates:  C:\Users\rlord\Downloads\neo4j-community-5.26.6-windows\neo4j-community-5.26.6\certificates
licenses:      C:\Users\rlord\Downloads\neo4j-community-5.26.6-windows\neo4j-community-5.26.6\licenses
run:           C:\Users\rlord\Downloads\neo4j-community-5.26.6-windows\neo4j-community-5.26.6\run
Starting Neo4j.
2025-05-25 05:11:00.881+0000 INFO  Logging config in use: File 'C:\Users\rlord\Downloads\neo4j-community-5.26.6-windows\
neo4j-community-5.26.6\conf\user-logs.xml'
2025-05-25 05:11:00.912+0000 INFO  Starting...
```



If you're using Neo4j Desktop, make sure that your local database is started. Open the Neo4j Browser – Localhost:7474, log in with your Neo4j username and password.

Test APOC and MongoDB Connections, in the browser's query box, run

Neo4j Graph Schema, each MongoDB document becomes a Person node, Properties are mapped directly, and relationships can be added later for more advanced graph queries.

```
C:\Users\rlord>C:\Users\rlord\Downloads\neo4j-community-5.26.6-windows\neo4j-community-5.26.6\bin\neo4j.bat console
```

Installing mongo-connector and neo4j_doc_manager

Neo4j's APOC library provides direct procedures to connect the MongoDB and import data, enabling seamless integration for analytics and graph modelling. This approach can be used as an alternative or complement to DocManager-based synchronization.

```
mongoc x Comm x Comm x mongoc x + v
C:\Users\rldord>pip install mongo-connector
Collecting mongo-connector
  Downloading mongo_connector-3.1.1-py2.py3-none-any.whl.metadata
  (60 kB)
Collecting pymongo>=2.9 (from mongo-connector)
  Downloading pymongo-4.13.0-cp312-cp312-win_amd64.whl.metadata
  (27 kB)
Collecting importlib-metadata>=0.6 (from mongo-connector)
  Downloading importlib_metadata-8.7.0-py3-none-any.whl.metadata
  (27 kB)
Collecting autocommand (from mongo-connector)
  Downloading autocommand-2.2.2-py3-none-any.whl.metadata (15 kB)
Collecting importlib-resources (from mongo-connector)
  Downloading importlib_resources-6.5.2-py3-none-any.whl.metadata
  (27 kB)
Collecting zipp>=3.20 (from importlib-metadata>=0.6->mongo-conn
  Downloading zipp-3.21.0-py3-none-any.whl.metadata (3.7 kB)
Collecting dnspython<3.0.0,>=1.16.0 (from pymongo>=2.9->mongo-c
  Downloading dnspython-2.7.0-py3-none-any.whl.metadata (5.8 kB)
Downloading mongo_connector-3.1.1-py2.py3-none-any.whl (60 kB)
Downloading importlib_metadata-8.7.0-py3-none-any.whl (27 kB)
Downloading pymongo-4.13.0-cp312-cp312-win_amd64.whl (898 kB)
898.7/898.7 kB 5.8 MB/s eta
Downloading autocommand-2.2.2-py3-none-any.whl (19 kB)
```

```
C:\Users\rldord>mongo-connector -m localhost:27017 -t localhost:7474 -d
neo4j_doc_manager
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "C:\Users\rldord\AppData\Local\Programs\Python\Python312\Scripts
\mongo-connector.exe\__main__.py", line 4, in <module>
  File "C:\Users\rldord\AppData\Local\Programs\Python\Python312\Lib\sit
e-packages\mongo_connector\connector.py", line 38, in <module>
    from mongo_connector.command_helper import CommandHelper
  File "C:\Users\rldord\AppData\Local\Programs\Python\Python312\Lib\sit
e-packages\mongo_connector\command_helper.py", line 20, in <module>
    from mongo_connector.namespace_config import NamespaceConfig
  File "C:\Users\rldord\AppData\Local\Programs\Python\Python312\Lib\sit
e-packages\mongo_connector\namespace_config.py", line 18, in <module>
    from collections import namedtuple, MutableSet
ImportError: cannot import name 'MutableSet' from 'collections' (C:\Us
ers\rldord\AppData\Local\Programs\Python\Python312\Lib\collections\__in
it__.py)
```

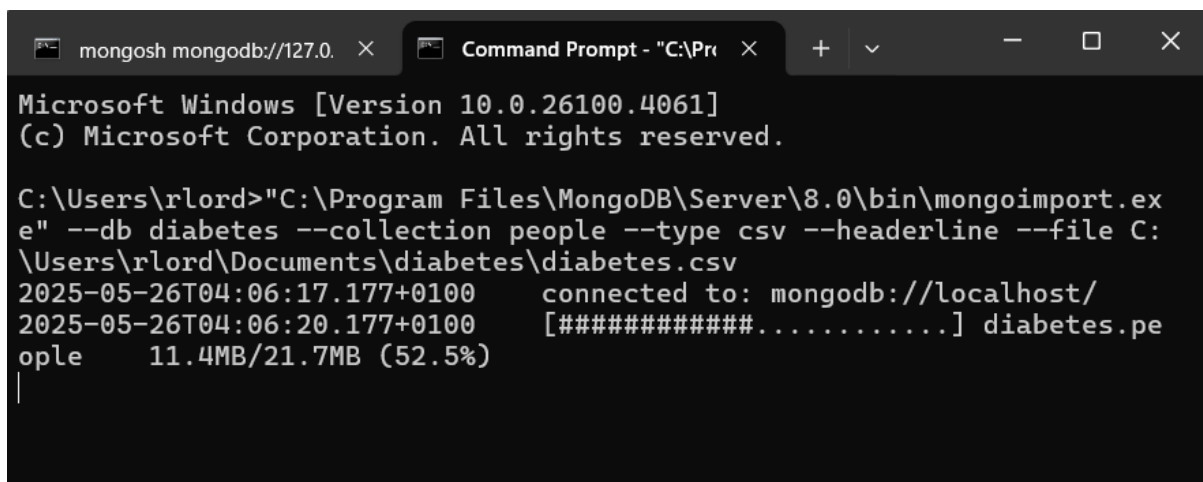
```
apoc.mongodb.enabled=true
```

```
apoc.mongodb.connection.mongo1.uri=mongodb://localhost:27017
```

Setting up the Database and Collection

First create a new MongoDB database called and a collection. Import the records from the json file. This was done using the mongoimport aswell. After importing its good to run a command just to double check that everything was imported correctly

```
C:\Users\rlord>"C:\Program Files\MongoDB\Server\8.0\bin\mongod.exe" --dbpath "C:\data\db"
{"t":{"$date":"2025-05-25T05:36:57.770+01:00"},"s":"I", "c":"CONTROL", "id":23285, "ctx":"thread1","msg":"Automatically disabling TLS 1.0, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2025-05-25T05:36:58.826+01:00"},"s":"I", "c":"CONTROL", "id":5945603, "ctx":"thread1","msg":"Multi threading initialized"}
{"t":{"$date":"2025-05-25T05:36:58.827+01:00"},"s":"I", "c":"NETWORK", "id":4648601, "ctx":"thread1","msg":"Implicit TCP FastOpen unavailable. If TCP FastOpen is required, set at least one of the related parameters","attr":{"relatedParameters":["tcpFastOpenServer","tcpFastOpenClient","tcpFastOpenQueueSize"]}}
{"t":{"$date":"2025-05-25T05:36:58.829+01:00"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"thread1","msg":"Initialized wire specification","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":25},"incomingInternalClient":{"minWireVersion":0,"maxWireVersion":25},"outgoing":{"minWireVersion":6,"maxWireVersion":25},"isInternalClient":true}}}
{"t":{"$date":"2025-05-25T05:36:58.830+01:00"},"s":"I", "c":"TENANT_M", "id":7091600, "ctx":"thread1","msg":"Starting TenantMigrationAccessBlockerRegistry"}
{"t":{"$date":"2025-05-25T05:36:58.830+01:00"},"s":"I", "c":"CONTROL", "id":4615611, "ctx":"initandlisten","msg":"MongoDB starting","attr":{"pid":27452,"port":27017,"dbPath":"C:/data/db","architecture":"64-bit","host":"RL-w"}}
```



The screenshot shows a Windows Command Prompt window with the following content:

```
mongosh mongodb://127.0. x Command Prompt - "C:\Pr x + v - □ x
Microsoft Windows [Version 10.0.26100.4061]
(c) Microsoft Corporation. All rights reserved.

C:\Users\rlord>"C:\Program Files\MongoDB\Server\8.0\bin\mongoimport.exe" --db diabetes --collection people --type csv --headerline --file C:\Users\rlord\Documents\diabetes\diabetes.csv
2025-05-26T04:06:17.177+0100 connected to: mongodb://localhost/
2025-05-26T04:06:20.177+0100 [#####.....] diabetes.people 11.4MB/21.7MB (52.5%)
|
```

```
C:\Users\rlord>"C:\Program Files\MongoDB\Server\8.0\bin\mongod.exe" --port 1000
{"t":{"$date":"2025-05-25T06:37:56.028+01:00"},"s":"I", "c":"CONTROL", "id":23285, "ctx":"thread1","msg":"Automatically disabling TLS 1.0, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2025-05-25T06:37:57.099+01:00"},"s":"I", "c":"CONTROL", "id":5945603, "ctx":"thread1","msg":"Multi threading initialized"}
```

Installing and Configuring Neo4j with APOC

Install Neo4j and the APOC plugin, then enable APOC procedures in the configuration. APOC is required for Neo4j to connect to external data sources like MongoDB and perform advanced operation.

Configuring APOC to Connect to MongoDB

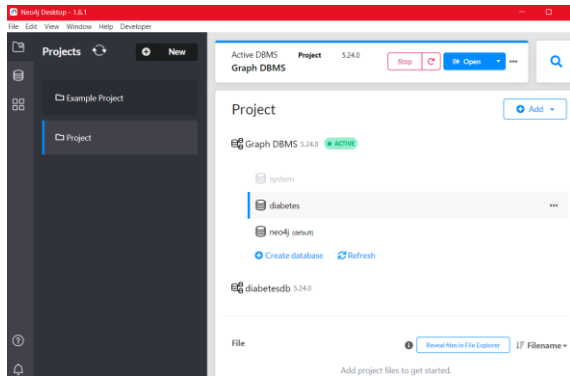
Set up the MongoDB connection details in Neo4j's configuration file. This enables Neo4j to securely access and import data directly from your MongoDB cluster.

Importing Data from MongoDB to Neo4j

Use APOC procedures to pull and transform data from MongoDB into Neo4j as nodes and properties this transfer your diabetes data into Neo4j, making it available for graph-based analysis and queries

Designing the Neo4j Graph Schema

Define how the imported data will be structured, nodes, labels, relationships in Neo4j. A well-designed schema makes complex queries easier and more efficient, matching your analysis needs.



Testing the Connection with the Diabetes Cluster

By running basic queries to be sure that Neo4j can access and display data from MongoDB. This helps verify that the integration is working and that the data transfer was successful.

```
mongosh mongodb://127.0.0.1:27027/
0
diabetes> use diabetes
switched to db diabetes
diabetes> db.people.findOne()
{
  _id: ObjectId('6833daa9a20732ceaf3091c5'),
  Diabetes_012: 0,
  HighBP: 0,
  HighChol: 0,
  CholCheck: 0,
  BMI: 25,
  Smoker: 1,
  Stroke: 0,
  HeartDiseaseorAttack: 0,
  PhysActivity: 1,
  Fruits: 0,
  Veggies: 0,
  HvyAlcoholConsump: 0,
  AnyHealthcare: 0,
  NoDocbcCost: 1,
  GenHlth: 3,
  MentHlth: 0,
  PhysHlth: 0,
  DiffWalk: 0,
  Sex: 0,
  Age: 7,
  Education: 6,
  Income: 1
}
diabetes>
```


Queries

1. Pregnant Young Women with Healthy Lifestyles

Find a women younger than 30, who is pregnant with a healthy lifestyle. The query below looks for Person nodes where, the person is female under 30 has is_pregnant set to true and either has a Healthy Diet or Regular Exercise, If a person has all of these it will return the persons name, age and lifestyle

```
match (p:Person) where p.gender = 'Female' and p.age < 30 and
p.is_pregnant = true and ('Healthy Diet' in p.lifestyle or 'Regular
Exercise' in p.lifestyle) return p.name, p.age, p.lifestyle
```

2. Obese Male Pensioners with Poor Health and Low Income

Looks for a person where gender is male, obesity is true, age is greater than 65, has a health condition or poor health, has low income.

```
match (p:Person)-[:HAS_CONDITION]→(h:HealthCondition) where p.gender =
'Male' and p.age ≥ 65 and (h.name = 'Obesity' or h.name = 'Poor
Health') and p.income = 'Low' return p.name, p.age, h.name AS
health_condition, p.income
```

3. Smokers with Diabetes and Multiple Risk Factors

Looks for a person where is_smoker is true, they have Diabetes, and they have multiple conditions like Obesity, Poor Diet

```
match (p:Person)-[:HAS_CONDITION]→(h:HealthCondition) where p.is_smoker
= true and 'Diabetes' IN h.name and (h.name = 'Obesity' or h.name =
'Hypertension' or h.name = 'Poor Diet') return p.name, p.age, h.name as
health_condition
```